

Inflation Inequality Across the Euro Area

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| What We Do

- ▶ Inflation inequality across households increases when relative-price shocks interact with heterogeneous household baskets.
- ▶ We construct monthly indices by income, age, and residence for all 20 euro-area countries, calibrated to official HICP weights.
- ▶ We simulate the effects of temporary VAT cuts, gas and electricity price measures, and fuel rebates on average inflation and inflation inequality.

Our contribution

1. A comprehensive framework to assess inflation inequality.
2. We connect household inflation exposure to price policies, quantifying their effects on both average inflation and income-group gaps.

Inflation Inequality and Policy Effects, 2021–2023

- ▶ Euro-area Q1–Q5 inflation gap rose to **0.9 point**, from 0.1 point in 2021, and returned to zero in 2023.
- ▶ Group averages conceal substantial within-category heterogeneity: households in the same income quintile can face markedly different inflation rates.
- ▶ Household energy contributed **1.6 points** to the 2022 Q1–Q5 gap; fuel, transport, and other products offset most of this exposure.
- ▶ Household price policies lowered annual inflation by **1.3 points in 2022** and **0.3 point in 2023**, while narrowing the Q1–Q5 gap by **0.4 point in both years**.

Linking Official Prices to Household Baskets

- ▶ We link monthly Eurostat HICP prices and official item weights to HBS expenditure shares through a harmonized COICOP bridge.
- ▶ Public HBS data provide consumption baskets by income, age, and residence; microdata add household expenditures, disposable income, and demographics.
- ▶ We use RAS to calculate expenditure weights that simultaneously satisfy two margins: the expenditure share of each household category and the official HICP weight of each product.
- ▶ Interpretation: baskets are held fixed between HBS waves, so the baseline abstracts from substitution and within-group heterogeneity (behavioral responses are not taken into account).

Different Consumption Baskets, Different Inflation Exposure

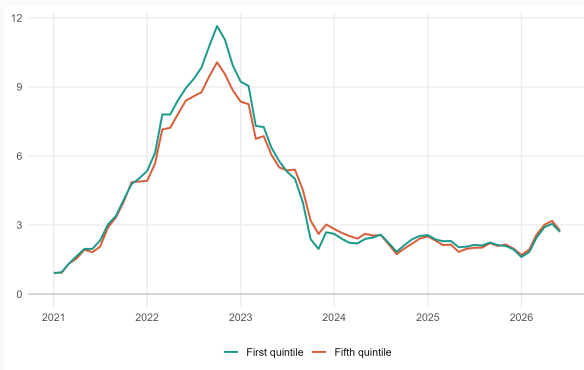
Effect of a 40% energy price shock

Expenditure share	First quintile of income	Fifth quintile of income	Inflation gap (Q1–Q5)
Housing Energy	10%	6%	–
Other consumption	90%	94%	–
Inflation if housing energy prices rise by 40%	4.0%	2.4%	1.6 pp

Illustrative example; other prices unchanged, fixed expenditure weights.

Low-Income Households Faced Higher Inflation During the Energy Shock

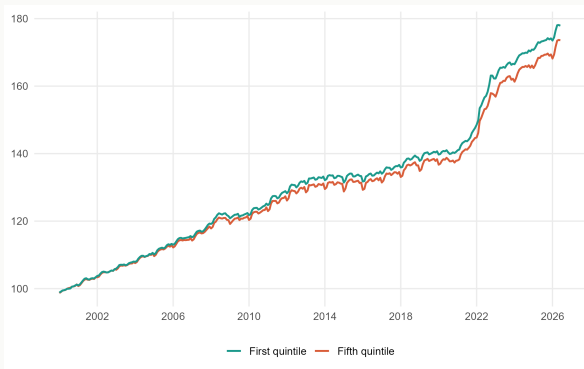
ECB analysis reports similar findings using a slightly different methodology (Bobasu et al., 2023).



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Since 2000, Only the 2021–23 Shock Had a Sizable Effect on Inflation Inequality

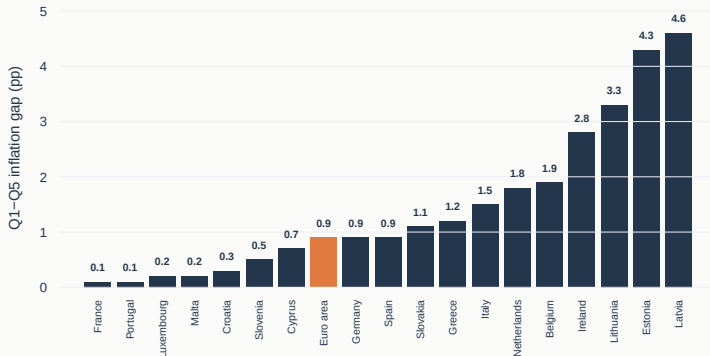
A return of the annual Q1–Q5 inflation gap to zero does not imply that the cumulative price-level gap has disappeared.



Source: Eurostat HICP and HBS; authors' calculations. Units: price index, 2000=100.

Q1–Q5 Inflation Gaps Across Euro-Area Countries in 2022

The euro-area average masks substantial cross-country differences in income-group inflation gaps.



Source: Eurostat HICP and HBS; authors' calculations. Countries with a negative Q1–Q5 gap in 2022 are excluded; the euro-area aggregate is shown as a benchmark.

Household Energy Widened the Q1–Q5 Gap, While Service Items Partly Offset It

The distributional impact of inflation depends on which prices rise, as well as on the size of the aggregate shock.

Product group	Q1–Q5 inequality contribution		
	2021	2022	2023
Food and non-alcoholic beverages	0.1	0.6	0.6
Alcohol, tobacco and narcotics	0.1	0.1	0.2
Clothing and footwear	0.0	0.0	-0.1
Housing (rentals and repairs)	0.0	-0.2	-0.1
Housing energy	0.4	1.6	0.1
Operation of personal transport equipment	-0.2	-0.4	0.0
Transport (excluding vehicle operation)	-0.1	-0.3	-0.2
Other	-0.1	-0.5	-0.5
Total	0.1	0.9	0.0

Source: Eurostat HICP and HBS; authors' calculations. Product rows are contributions to the Q1–Q5 annual inflation-rate gap, in percentage points.

Three Instruments Lowered Retail Energy Prices

The counterfactual covers **100 price-policy measures** (prices held below market levels or tax cuts) across all 20 euro-area countries, costing **EUR 268.9 billion**; **88%** of the cost was untargeted.

- ▶ **Price caps and tariff shields** limited household electricity and gas prices.
France: gas tariffs were frozen and the 2022 electricity increase was capped at 4%.
- ▶ **Temporary VAT and excise-tax reductions** lowered consumer energy prices.
- ▶ **Fuel rebates and regulated discounts** reduced prices at the pump.
France: the fuel rebate rose from EUR 0.18 to EUR 0.30 per litre in autumn 2022.

Source: National policy sources; Sgaravatti et al.; authors' calculations.

Defining the Price-Policy Counterfactual

CF = counterfactual; O = observed; TRV_{gas} : gas tariff (EUR/kWh); TRV_e : electricity tariff (EUR/kWh).

1. Gas | Nov. 2021–June 2023

$$\Delta P_{\text{gas},t}^{CF} = \beta_{\text{gas}} \Delta TRV_{\text{gas},t}^{CF}$$

unit-free decimal rates.

$$\beta_{\text{gas}} = 0.82.$$

3. Motor fuels | Apr.–Dec. 2022

$$p_{\text{fuel},t}^{CF} = p_{\text{fuel},t}^O + r_t \quad (\text{EUR/litre, incl. taxes})$$

Chained index:

$$P_{\text{fuel},t}^{CF} = P_{\text{fuel},t-1}^{CF} \frac{p_{\text{fuel},t}^O + r_t}{p_{\text{fuel},t-1}^O + r_{t-1}}.$$

Monthly rebates (EUR/litre):

Apr.–Aug.	$r = 0.18$	Sept.–Oct.	$r = 0.30$
November	$r = 0.22$	December	$r = 0.10$

2. Electricity | Feb. 2022–Feb. 2024

$$\Delta P_{e,t}^{CF} = \beta_e \Delta TRV_{e,t}^{CF}$$

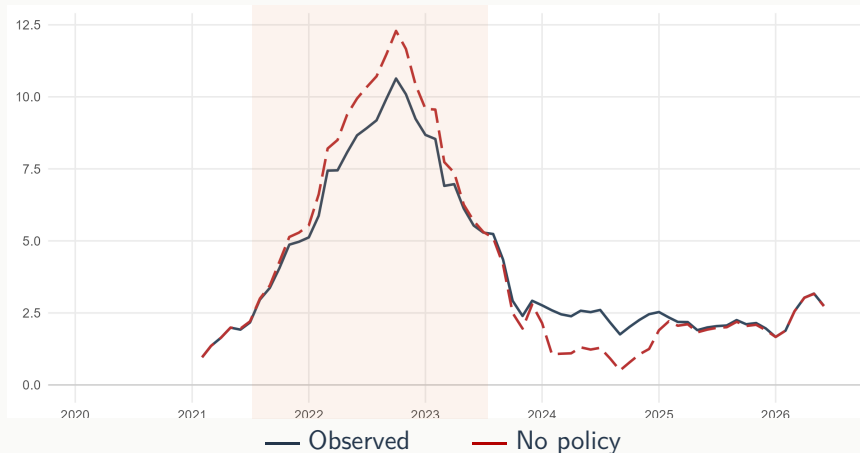
unit-free decimal rates.

$$\beta_e = 1.02.$$

Date	Observed reference	$\Delta TRV_{e,t}^{CF}$
Feb. 2022	Jan. 2022	33.24%
Feb. 2023	Jan. 2023	108.91%
Feb. 2024	Feb. 2024	0% [†]

[†]Model closure, not a tariff estimate: $P_{e,t}^{CF} = P_{e,t}^O$.

Price Policies Compressed the Inflation Peak



Source: Eurostat and national statistical institutes; authors' calculations. Units: year-on-year inflation, in percent; January 2021–July 2026.

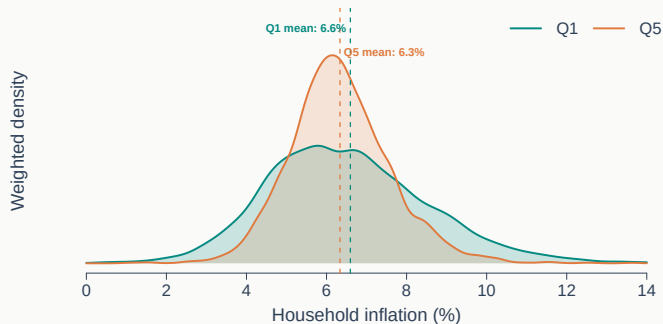
Policy Effects on the Q1–Q5 Gap

Price measures can narrow the Q1–Q5 gap without income targeting: electricity and gas support reduced the gap, while fuel support partly offset this effect in 2022.



Source: Eurostat and national statistical institutes; authors' calculations. Annual-average Q1–Q5 inflation-rate gaps, in percentage points.

Most Inflation Dispersion Is Within Income Quintiles



$$\text{Var}(\pi_h) = 2.9738$$

$$\underbrace{0.0202}_{\text{between quintiles}} + \underbrace{2.9535}_{\text{within quintiles}}$$

Source: French HBS and HICP, 2022. Total-income quintiles; rents excluded. Authors' calculations.

Older Households Faced Higher Inflation



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Price Policies Reduced Average Inflation and Inflation Inequality

- ▶ The inflation shock affected households very differently, even at a given income level: most inflation dispersion occurs **within** income groups.
- ▶ Q1–Q5 inflation inequality rose from **0.1 point in 2021** to **0.9 point in 2022**, then returned to zero in 2023; national experiences remained far more dispersed.
- ▶ Inflation exposure varied substantially **across countries**.
- ▶ Price policies reduced annual inflation by **1.3 points in 2022** and **0.3 point in 2023**; they reduced inequality by **0.4 point in both years**.
- ▶ Administered household-energy prices generated most of the distributional cushioning; fuel support was mildly regressive in 2022.
- ▶ Our policy counterfactuals quantify effects on consumer prices and inflation gaps, rather than a complete welfare assessment.

Appendix

Defining the Price-Policy Counterfactual

- ▶ The counterfactual removes these price-based interventions component by component while retaining observed prices for unaffected products.

Example: reversing a temporary VAT cut

$$\tilde{p}_{j,t} = p_{j,t} \frac{1 + \tau_{j,t}^0}{1 + \tau_{j,t}^1}, \quad \tau_{j,t}^0 = \text{VAT rate}, \quad \tau_{j,t}^1 = \text{temporary VAT rate}.$$

Belgium: cutting VAT from 21% to 6% implies

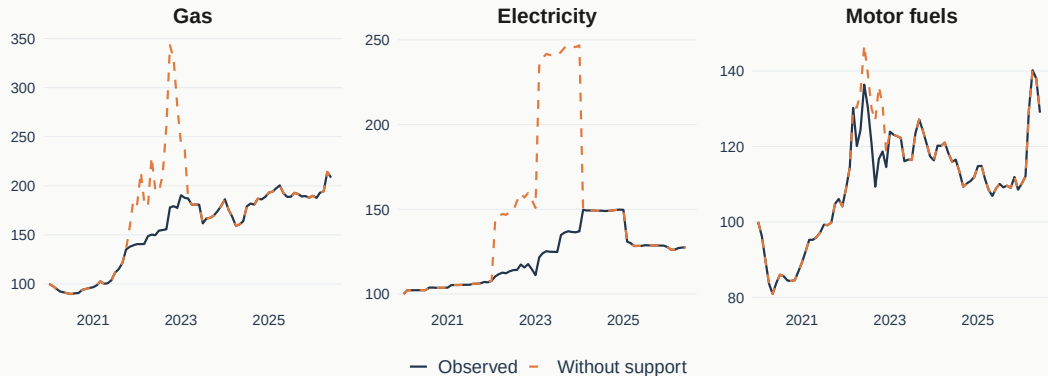
$$\tilde{p}_{j,t} = p_{j,t} \frac{1.21}{1.06} \simeq 1.142 p_{j,t}.$$

j = electricity and t = March 2022–December 2023.

Source: National policy sources, Sgaravatti et al.; authors' calculations.

France: Observed and Counterfactual Energy Price Indices

January 2020 = 100; January 2020–June 2026. Vertical scales differ across panels.



Source: French observed and no-policy component price series; authors' calculations.

Household Weights Combine HBS and HICP

Fixed 2020 expenditures; annual official HICP weights; actual and imputed rents excluded.

$$\omega_{j,ic,y} = \frac{\omega_{j,ic,2020}^{\text{HBS}} \omega_{j,c,y}^{\text{HICP}}}{\sum_{k \in \mathcal{J}_{c,y}} \omega_{k,ic,2020}^{\text{HBS}} \omega_{k,c,y}^{\text{HICP}}}.$$

$$P_{ic,y,m} = \sum_{j \in \mathcal{J}_{c,y}} \omega_{j,ic,y} \frac{p_{c,j,y,m}}{p_{c,j,y-1,12}}.$$

Policy Effects Differed Sharply Across Countries and Years

Country	Average-inflation effect			Q1–Q5 inequality effect		
	2021	2022	2023	2021	2022	2023
Germany	0.0	-0.5	-0.7	0.0	-0.2	-0.7
France	-0.1	-2.2	-0.5	0.0	-0.9	-0.4
Italy	-0.2	-1.3	0.7	-0.1	-0.1	-0.1
Spain	-0.4	-2.0	-0.4	-0.2	-0.7	-0.4
Netherlands	0.0	-1.6	-0.7	0.0	-0.5	-0.6
Portugal	0.0	-0.8	-1.4	0.0	-0.3	-0.9
Slovakia	0.0	0.0	-3.0	0.0	0.0	-1.9
Euro area	-0.1	-1.3	-0.3	-0.1	-0.4	-0.4

- Negative effects indicate that unconventional fiscal policies reduced inflation or the Q1–Q5 gap.

Source: Eurostat and national statistical institutes; authors' calculations. Entries are observed minus no-policy annual-average inflation rates, in percentage points.

Rural Households Faced Higher Inflation at the Peak



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Energy Import Prices Mainly Raise Average Inflation

Product group	Average inflation	Q1–Q5 gap	Direct Q1–Q5 gap	Indirect Q1–Q5 gap
Food and non-alcoholic beverages	0.5	0.1	0.0	0.1
Alcohol, tobacco and narcotics	0.1	0.1	0.0	0.1
Clothing and footwear	0.1	0.0	0.0	0.0
Housing (rentals and repairs)	0.2	0.0	0.0	0.0
Housing energy	1.6	0.7	0.2	0.5
Vehicle operation	1.7	-0.6	-0.1	-0.6
Transport (excluding vehicle operation)	1.1	-0.1	0.0	-0.1
Other	0.7	-0.2	0.0	-0.1
Total	6.0	-0.1	0.0	-0.1

- ▶ Housing energy widens the Q1–Q5 gap by **0.7 point**; other products offset this effect.

Source: Eurostat trade statistics, HBS and FIGARO; authors' calculations. Observed extra-EU energy import-price shock: 83.9% in 2022. Effects in pp; components sum before rounding.

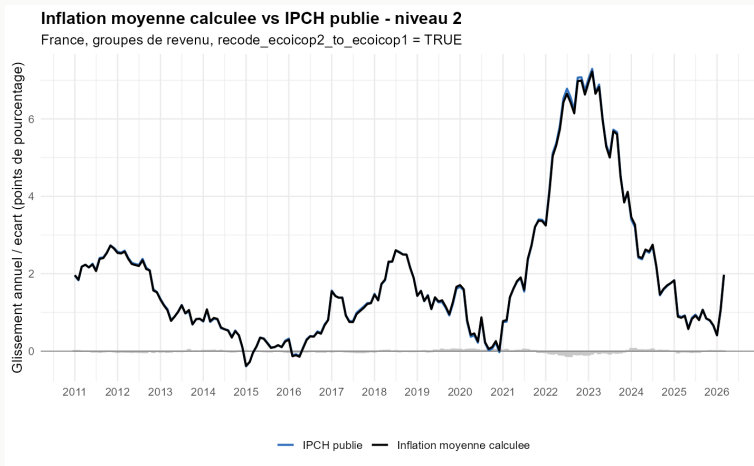
Household-Energy Measures Account for the Inequality Reduction

Product group	Average-inflation effect			Q1–Q5 inequality effect		
	2021	2022	2023	2021	2022	2023
Food and non-alcoholic beverages	0.0	0.0	-0.1	0.0	0.0	0.0
Housing energy	-0.1	-1.0	-0.5	-0.1	-0.5	-0.3
Fuel	0.0	-0.3	0.3	0.0	0.1	0.0
All other products	0.0	0.0	0.0	0.0	0.0	0.0
Total	-0.1	-1.3	-0.3	-0.1	-0.4	-0.4

- ▶ Household energy lowers the Q1–Q5 gap by **0.5 point in 2022** and **0.3 point in 2023**; the 2022 fuel effect offsets 0.1 point.

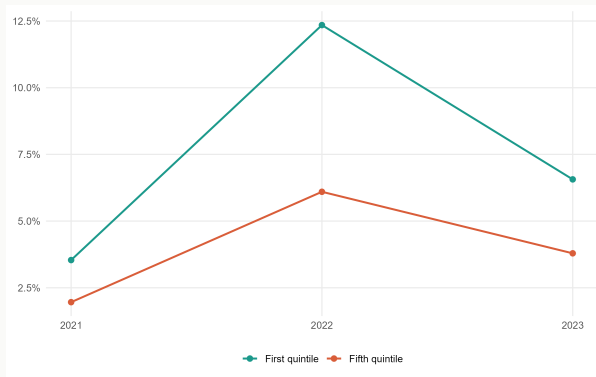
Source: Eurostat and national statistical institutes; authors' calculations. Product contributions to annual policy effects, in percentage points.

RAS-Calibrated Inflation Tracks the Published HICP



Source: INSEE, Eurostat; authors' calculations. Units: year-on-year inflation, in percent.

Inflation Burden Is Larger for Low-Income Households



- ▶ The Q1 burden reached **3.5%**, **12.3%**, and **6.6%** of net income in 2021–2023, versus **2.0%**, **6.1%**, and **3.8%** for Q5.

Source: Eurostat HICP and HBS microdata; authors' calculations. Units: annual inflation burden, in percent of household net income.

What Our Inflation Indices Capture

Channel	Mechanism	Covered here?
Consumption baskets	Different expenditure shares imply different inflation exposure	Yes
Nominal income	Wages, pensions and transfers adjust to inflation	No
Balance sheets	Inflation changes the real value of nominal assets and debt	No

Inflation Inequality Was Highly Uneven Across Countries

Country	2021	2022	2023	Country	2021	2022	2023
Austria	-0.5	-0.2	0.5	Ireland	0.4	2.8	1.1
Belgium	1.2	1.9	-2.3	Italy	0.1	1.5	0.4
Croatia	0.0	0.3	-0.3	Latvia	-0.2	4.6	2.4
Cyprus	0.0	0.7	0.2	Lithuania	0.2	3.3	0.4
Estonia	0.5	4.3	1.5	Luxembourg	0.1	0.2	-0.1
Finland	-0.2	-1.1	0.2	Malta	-0.4	0.2	1.2
France	0.3	0.1	0.1	Netherlands	-0.3	1.8	-0.9
Germany	-0.4	0.9	0.4	Portugal	0.7	0.1	-0.9
Greece	0.3	1.2	-0.5	Slovakia	-0.9	1.1	1.3
Spain	0.8	0.9	-0.7	Slovenia	0.1	0.5	1.4
Euro area					0.1	0.9	0.0

Source: Eurostat HICP and HBS; authors' calculations. Units: Q1 minus Q5 annual-average inflation rates, in percentage points.

Administered Prices Drove the Distributional Effect

Policy category	Fiscal cost EUR bn	Average-inflation effect			Q1–Q5 inequality effect		
		2021	2022	2023	2021	2022	2023
VAT cuts	63.3	0.0	-0.1	-0.1	0.0	0.0	0.0
Administered prices	163.8	0.0	-0.5	-0.8	0.0	-0.3	-0.5
Fuel subsidies and rebates	17.0	0.0	-0.3	0.3	0.0	0.1	-0.1
Other calibrated layer	16.3	0.0	-0.3	0.1	0.0	-0.2	0.0
Unallocated costs	8.5	–	–	–	–	–	–
Decomposition total	268.9	-0.1	-1.2	-0.5	-0.1	-0.4	-0.5
Untargeted measures	236.2	-0.1	-0.8	-0.1	0.0	-0.2	-0.3
Targeted measures	32.8	0.0	-0.5	-0.4	0.0	-0.2	-0.2
Main-paper joint effect	268.9	-0.1	-1.3	-0.3	-0.1	-0.4	-0.4

The decomposition has not yet been reconciled with the main-paper joint effect.

Source: Bruegel; national sources; authors' calculations. Annual policy effects are observed minus no-policy rates, in percentage points. Category effects need not sum because policies interact.

French Energy Price Shields: Comparing Evaluations

Inflation estimates differ across models; distributional outcomes are not directly comparable.

Institute	Model	Aggregate inflation effect		Inflation-gap effect	
		2022	2023	2022	2023
This paper	Price counterfactuals	−2.2 pp	−0.5 pp	−0.9 pp	−0.4 pp
Banque de France	FR-BDF	−2.6 pp	+0.4 pp	NR	NR
OFCE / CAE	ThreeME; general equilibrium	−0.73 pp	−0.68 pp	NR	NR
CEPREMAP	CepreHANK	−1.1 pp	−1.8 pp	NR	NR

NR: not reported in the cited study. Policy coverage and inflation conventions differ.

Source: This paper, France (slide 21); BdF (2024); OFCE/CAE (2023); CEPREMAP (2022).

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